WORKED SOLUTION

STEP II 1995, Question 13

Mechanics • Poisson distribution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q13

The key idea

Despite the database tag, this is a probability question dressed up as business: the no-shows are a binomial count with a large number of trials and a tiny success probability, which is exactly when the Poisson distribution takes over. After that, the trick for optimising over a whole number k is not calculus but the *first difference* $u_{k+1} - u_k$: as long as selling one more ticket increases the expected profit, keep going. Once that difference is shown to be decreasing, the best k is where it first stops being positive.

Solution

The Poisson model

Suppose each of the $N + k$ ticket holders independently fails to turn up with probability ϵ. Then the number R who fail to turn up has the binomial distribution $\mathrm{B}(N+k, \epsilon)$, with mean $(N+k)\epsilon$. Since k is very small compared with N, this mean is approximately $N\epsilon = \lambda$. A binomial distribution with a large number of trials and a very small success probability is well approximated by a Poisson distribution with the same mean: for fixed r, as $n \to \infty$ with $n\epsilon = \lambda$ fixed,

$$\binom{n}{r}\epsilon^r(1-\epsilon)^{n-r} = \frac{n(n-1)\cdots(n-r+1)}{n^r}\cdot\frac{\lambda^r}{r!}\left(1-\frac{\lambda}{n}\right)^{n-r} \;\to\; \frac{\lambda^r}{r!}\,\mathrm{e}^{-\lambda},$$

because the first factor tends to 1 and $(1-\lambda/n)^n \to \mathrm{e}^{-\lambda}$. So, approximately, $R \sim \mathrm{Po}(\lambda)$. From now on we take this as exact and write

$$p_r = \mathrm{P}(R = r) = \frac{\lambda^r}{r!}\,\mathrm{e}^{-\lambda}.$$

The profit

The airline sells $N + k$ tickets at £A profit each, which gives $A(N+k)$ whatever happens. If r passengers fail to turn up, then $N + k - r$ arrive for N seats.

- If $r \geq k$, at most N people arrive and everyone flies: nobody is compensated, and $\max(0, k-r) = 0$.
- If $r < k$, then $k-r$ people cannot be flown, each costing £B, and $\max(0, k-r) = k-r$.

In both cases the profit in pounds is $A(N+k) - B\max(0, k-r)$.

The expected profits $u_0, \ldots, u_3$

Taking expectations, only $r = 0, 1, \ldots, k-1$ contribute to the penalty, so

$$u_k = A(N+k) - B\sum_{r=0}^{k-1}(k-r)\,p_r.$$

With $p_0 = \mathrm{e}^{-\lambda}$, $p_1 = \lambda\mathrm{e}^{-\lambda}$ and $p_2 = \frac{1}{2}\lambda^2\mathrm{e}^{-\lambda}$:

$$\begin{aligned} u_0 &= AN, \\ u_1 &= A(N+1) - B\mathrm{e}^{-\lambda}, \\ u_2 &= A(N+2) - B\mathrm{e}^{-\lambda}(2+\lambda), \\ u_3 &= A(N+3) - B\mathrm{e}^{-\lambda}\left(3 + 2\lambda + \tfrac{1}{2}\lambda^2\right). \end{aligned}$$

The differences v_k

Subtracting the general formulae,

$$\begin{aligned} v_k = u_{k+1} - u_k &= A - B\Big[\sum_{r=0}^{k}(k+1-r)\,p_r - \sum_{r=0}^{k-1}(k-r)\,p_r\Big] \\ &= A - B\Big[p_k + \sum_{r=0}^{k-1}\big((k+1-r) - (k-r)\big)p_r\Big] \\ &= A - B\sum_{r=0}^{k} p_r, \end{aligned}$$

where the $r = k$ term of the first sum, $(k+1-k)p_k = p_k$, has no partner in the second sum. In words, $v_k = A - B\,\mathrm{P}(R \leq k)$. This makes sense: the $(k+1)$th extra ticket earns £A for sure, and it costs an extra £B exactly when that passenger ends up bumped, which happens when $R \leq k$.

Ask yourself Why does the bracket collapse to a plain sum of probabilities?

For each $r \leq k-1$, the coefficient of p_r goes up by exactly one when k increases by one (one more bumped passenger in that outcome), and the outcome $r = k$ newly starts to cost one payment. So every r from 0 to k contributes p_r once.

Then

$$v_k - v_{k+1} = B\,p_{k+1} = B\,\frac{\lambda^{k+1}}{(k+1)!}\,\mathrm{e}^{-\lambda} > 0,$$

since $B > 0$ and $\lambda > 0$. So $v_k > v_{k+1}$ for every k: the differences strictly decrease.

Finally, $\sum_{r=0}^{k} p_r$ is a partial sum of $e^{-\lambda}\sum_{r\geq 0}\lambda^r/r! = e^{-\lambda}e^{\lambda} = 1$, so it tends to 1 as $k \to \infty$, and

$$v_k \to A - B \qquad \text{as } k \to \infty,$$

as required. Since $B > A$, this limit is negative.

Advice to Fly By Night

Write $F(k) = \mathrm{P}(R \leq k) = \sum_{r=0}^{k} p_r$, so $v_k = A - BF(k)$, and

$$u_{k+1} > u_k \iff v_k > 0 \iff F(k) < \frac{A}{B}.$$

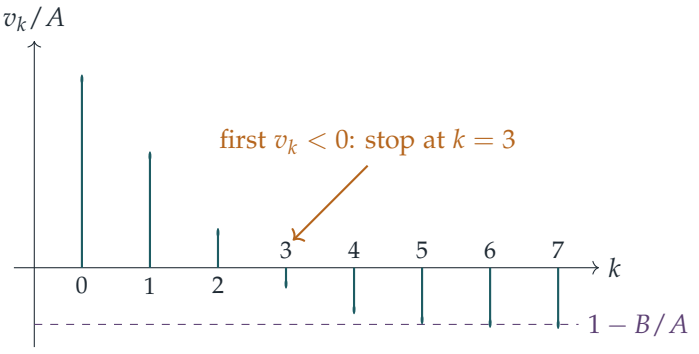
Because v_k is strictly decreasing with a negative limit, the sequence $v_0, v_1, v_2, \ldots$ is positive up to some point and negative (or zero once) from then on. So u_k rises and then falls, and its maximum is at the first k for which the next ticket no longer pays:

$$\boxed{\text{choose the smallest } k \text{ with } F(k) = \sum_{r=0}^{k} \frac{\lambda^r}{r!}e^{-\lambda} \geq \frac{A}{B}.}$$

In particular, if $e^{-\lambda} \geq A/B$ (very reliable passengers, or a very heavy penalty), then $v_0 \leq 0$ and the airline should not overbook at all: $k = 0$. If $F(k) = A/B$ exactly, then $v_k = 0$ and k and $k+1$ give the same expected profit. Since $A/B < 1$ and $F(k) \to 1$, such a k always exists.

So the advice is: keep selling extra tickets while the probability that at most k people fail to turn up is less than the ratio A/B of ticket profit to compensation; stop at the first k where it reaches A/B.

For example, with $\lambda = 2$ and $B = 1.25A$ (so $A/B = 0.8$), we get $F(0) \approx 0.135$, $F(1) \approx 0.406$, $F(2) \approx 0.677$ and $F(3) \approx 0.857$, so the best choice is $k = 3$. The plot shows v_k/A for this case.



Checks

- From the general formula, $u_1 - u_0 = A - B\mathrm{e}^{-\lambda} = A - Bp_0$ and $u_2 - u_1 = A - B\mathrm{e}^{-\lambda}(1+\lambda) = A - B(p_0 + p_1)$, matching v_0 and v_1.
- Limiting cases: if $B \to A$ then $A/B \to 1$ and k grows without bound (bumping is almost free); if $B \to \infty$ then $A/B \to 0$ and $k = 0$.
- Numerically, for hundreds of random choices of N, A, B, λ we computed u_k directly from the profit formula and confirmed the formulae for $u_0, \ldots, u_3$ and v_k, that $v_k - v_{k+1} = Bp_{k+1}$, and that the k maximising u_k is the smallest k with $F(k) \geq A/B$. For $N = 400$, $k = 3$, $\epsilon = 0.01$, the binomial and Poisson probabilities of $r = 3$ no-shows are 0.194 and 0.195.

Common slips

- Including the $r \geq k$ terms in the penalty sum: they contribute nothing, since then everyone who turns up gets a seat.
- Differentiating u_k with respect to k. The variable is an integer, and the difference v_k is the right tool.
- Stating the answer as "$v_k = 0$". In general no k makes v_k exactly zero; the answer is the first k where $v_k \leq 0$.

Beyond STEP

The argument "the differences decrease, so stop where they change sign" is the discrete version of a concave function having its maximum where the derivative vanishes. The rule "stop when $\mathrm{P}(R \leq k)$ reaches A/B" is the classic *newsvendor* (critical fractile) rule from operations research, and airline overbooking is a standard application of it. The Poisson limit of the binomial, made rigorous above, is one of the first limit theorems in a university probability course. Grimmett and Welsh, *Probability: An Introduction*, covers both the Poisson approximation and expectations of functions like $\max(0, k - R)$.

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